08. Model Evaluation

Interpret the metrics and confusion matrices in the saved notebook.

Project: CreditWise – Loan Approval Prediction

Source: supplied Loan-Approval-main.zip | Notebook: credit_wise.ipynb | Dataset: loan_approval_data.csv

First-pass results

Model	Confusion matrix [[TN,FP],[FN,TP]]	Interpretation
Logistic Regression	[[126,13],[14,47]]	13 false approvals under positive=Yes; 14 missed approvals
KNN	[[120,19],[29,32]]	More false negatives; lower recall
GaussianNB	[[128,11],[16,45]]	Highest precision among first-pass models

Second-pass results

Model	Confusion matrix	Interpretation
Logistic Regression	[[125,14],[10,51]]	Highest recall and F1 in second pass
KNN	[[122,17],[26,35]]	Still lowest recall/F1
GaussianNB	[[129,10],[18,43]]	Highest precision; more false negatives

Metric meaning

- Precision = TP/(TP+FP): among applications predicted Yes, how many were actually Yes.
- Recall = TP/(TP+FN): among actual Yes applications, how many were identified.
- F1 = harmonic mean of precision and recall; useful when both matter.
- Accuracy = (TP+TN)/N: overall correctness, potentially misleading with imbalance or asymmetric costs.

Positive-class semantics

> Because LabelEncoder maps No to 0 and Yes to 1, the reported precision/recall/F1 are for class Yes (approval). If the business objective is to identify risky applications for rejection, the positive class and costs may need to be reframed; always state which class the metrics refer to.

Evaluation caveats

- Metrics are calculated on a single 20% split and can vary with split choice.
- No stratification, cross-validation, ROC-AUC, PR-AUC, calibration or threshold analysis is reported.
- The target's missing labels were imputed as No, so reported scores are based partly on fabricated outcomes.
- Pre-split imputation/encoding can leak information from the test partition.
- The dataset has only 1,000 rows; uncertainty intervals and external/temporal validation are important.

Which model is best?

If the immediate goal is to maximize approval-class precision, GaussianNB is best in both reported passes (0.804 then 0.811). If the goal balances precision and recall, second-pass Logistic Regression has the highest F1 (0.810), accuracy (0.880) and recall (0.836). There is no universal winner without business costs and validation.